ADSP 31014 Statistical Models for Data Science
Course Project Part 1 Example

Overview of Data Set

The data set consists of 6.38M taxi trips in the City of Chicago in year 2022. It was obtained from the Chicago Data Porter https://data.cityofchicago.org/Transportation/Taxi-Trips-2022/npd7-ywjz. Most of the taxi trips in the year were represented. Each trip consists of information about its start and end time, pick-up and drop-off location, along with trip duration, distance, fare and payment type, and individual taxi ID and company. To protect privacy, times are rounded to the nearest 15 minutes, and locations are mapped to coarser Census Tract or Community Area centroid coordinates.

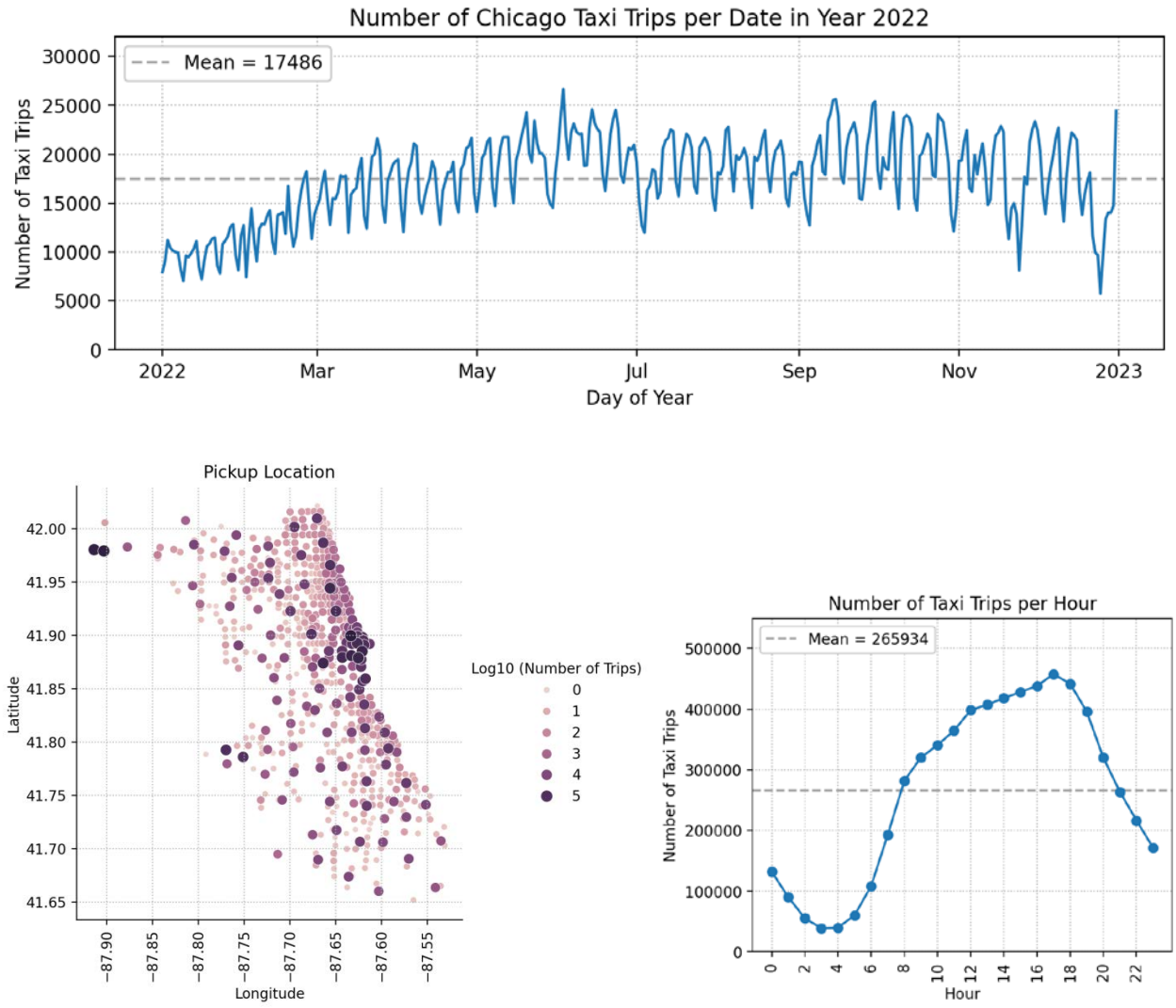


Figure 1: The number of taxi trips per date (top), hour (low right), and pickup location (low left).

Data Table Schema

The data set consists of 6,382,425 rows and 23 columns. Each row represents one taxi trip.

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Column Name	Description	Data Type	Example Value	Notes
Trip ID	Unique ID per trip	string	“4dd1d7066a400f3647bd944025a3316e5a27e4cb”	• Each row has unique Trip ID
Taxi ID	Unique ID per taxi	string	“5ee5354efef91805467615c7ecae65e68e119bd0f87a5650a6eb7f840aba79c76af183258958edf1599c2c308c6e5249b02b5cadb8c2c84d9fab89f38b5bd762”	• 2,950 unique taxi IDs • Most popular 100 taxi IDs appear about 5,000–10,000 times • 354 rows of missing data
Trip Start Timestamp	Trip start and end dates and times	string	“2022-05-09 14:15:00”	• Rounded to nearest 15 minutes • Had daylight saving • 122 rows of end timestamp before start timestamp • 212 rows of missing data
Trip End Timestamp		string	“2022-05-09 14:15:00”	
Trip Seconds	Trip duration (in seconds)	integer	370	• 155,667 rows of 0’s • 6,438 rows of 5–24 hours • Many values are multiples of 60’s despite in seconds • 1,465 rows of missing data
Trip Miles	Trip distance (in miles)	decimal	1.03	• 794,965 rows of 0’s • 226 rows of 200+ miles • All are only 2 decimal places • 56 rows of missing data
Pickup Census Tract	US Census Tract IDs of pick-up and drop-off locations	integer	17031081401	• Often missing for location outside Chicago • If too few trips per Census Tract, then it is missing and up-level to Community Area. See here. • 3.7M rows of missing Census Tract data • 0.5M+ rows of missing Community Area data
Dropoff Census Tract		integer	17031080100	
Pickup Community Area	Community Area IDs of pick-up and drop-off locations	integer	8	
Dropoff Community Area		integer	8	
Fare	Trip fare	decimal	6.5	• 6,150 rows of $0 fare • 874 rows of $500+ fare with 76% from one taxi company • Cash tip was usually not recorded • 3,536 rows of missing data • Cash payment and a few taxi companies have unusually large missing %
Tips	Tip for trip	decimal	3.0	
Tolls	Tolls for trip	decimal	0.0	
Extras	Extra charges	decimal	0.0	
Trip Total	Total cost of trip	decimal	10.0	
Payment Type	Type of payment	string	“Credit Card”	• Mostly “Cash” and “Credit Card” among 8 categories
Company	Taxi company	string	“Flash Cab”	• 38 companies but 4 seems typo
Pickup Centroid Latitude	Latitude and longitude of the centers of the Census Tract or Community Area of the	decimal	41.895033	• Pick-up (also drop-off) latitude / longitude / location are internally consistent • 0.5M+ rows of missing data
Pickup Centroid Longitude		decimal	-87.619711	
Pickup Centroid Location		string	“POINT (-87.6197106717 41.8950334495)”	
Dropoff Centroid		decimal	41.90752	

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Latitude	pick-up and drop-off locations			
Dropoff Centroid Longitude		decimal	-87.626659	
Dropoff Centroid Location		string	“POINT (-87.6266589003 41.9075200747)”	

Data Cleaning and Processing

The two main data quality issues are missing pick-up / drop-off locations and outliers in trip duration, distance, and fare. Missing data seems to be mostly missing at random.

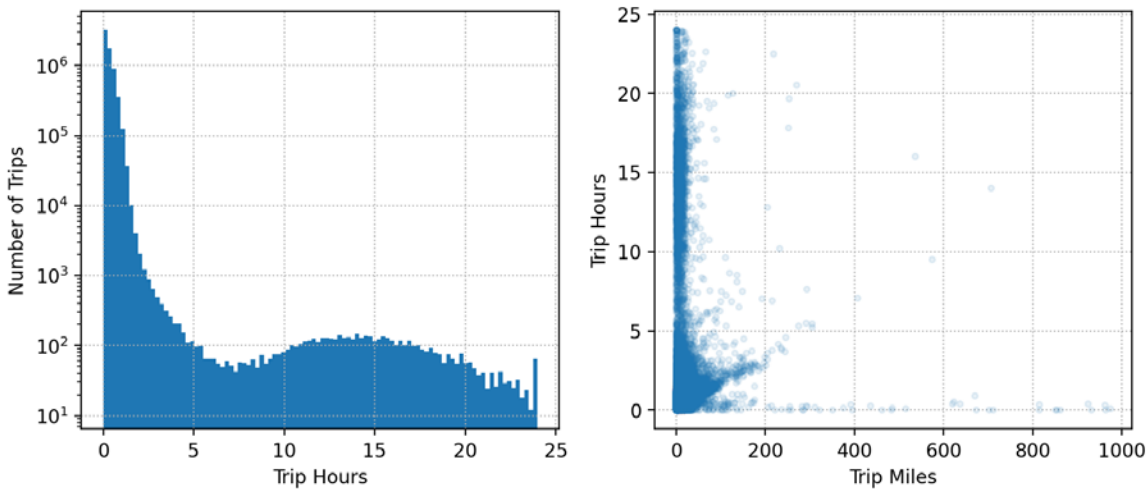


Figure 2: Outliers of trip duration (left) and the noisy impact to its relationship with trip distance (right).

Consequently, the data set is cleaned by removing rows of the following kinds:

- Row with any missing data
- Trip Start Timestamp > Trip End Timestamp
- Trip Seconds and the Trip Start / End Timestamp have discrepancy beyond 15 minutes
- Trip Seconds outside 1 seconds to 2 hours
- Trip Miles outside 0.01–100 miles
- Fare outside $3.25–$200 (since $3.25 is Chicago taxi base fare)
- Tips outside $0–$50
- Tolls outside $0–$20
- Extras outside $0–$20
- Payment Type is “Dispute” or “No Charge”

Moreover, 4 typos in taxi company name are fixed. Payment Type not of “Cash” and “Credit Card” are re-categorized as “Other”. The resulting cleaned data set has 4,769,009 rows.

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